

Distribution Narrowing Under Synthetic Data Contamination: A Multi-Seed Study on Linguistic Novelty Comprehension

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Abstract

We investigate whether training language models on synthetic-contaminated data degrades their ability to process genuinely novel linguistic constructions. Using GPT-2 (124M) fine-tuned on WikiText-103 across multiple generations with contamination ratios from 0% to 100%, we measure the *log-probability gap* between standard and novel text as a continuous proxy for distributional competence. All conditions are replicated with $n=3$ seeds and reported as mean $\pm$ standard deviation. Under shallow fine-tuning (500 steps), moderate contamination (10–50%) produces distribution *narrowing*—the model becomes more confident on standard text while novel-text processing is preserved—but not competence loss. Under deep fine-tuning (5,000 steps), the pretrained buffer breaks: the gap at 100% contamination widens to 2.30 ± 0.09 ($p < 0.001$), accompanied by perplexity collapse to 854 ± 23 by generation 1 and $1,482 \pm 30$ by generation 2. Generation diversity collapses even at low contamination levels, providing the most sensitive early warning of distribution narrowing. These results establish that pretrained representations offer a buffer against moderate contamination under light training, but this buffer is not durable: sufficient optimization pressure on contaminated data breaks through pretrained representations and produces genuine competence degradation.

1 Introduction

Over 50% of internet text may be machine-generated by 2026 [Hataya et al., 2023], and each new generation of language models risks training on the outputs of its predecessors. Shumailov et al. [2024] demonstrated that this recursive training loop causes *model collapse*: the tails of the original data distribution vanish irreversibly, and output quality degrades to repetitive, incoherent text. Alemohammad et al. [2023] formalized a complementary result—self-consuming generative models undergo “Model Autophagy Disorder,” progressively losing distributional diversity regardless of architecture.

This paper asks a stronger question: does synthetic contamination destroy a model’s ability to *comprehend* genuinely novel linguistic constructions—not just degrade its outputs, but eliminate the representational capacity for out-of-distribution language? If so, the consequence is not merely worse generation quality but a structural competence ceiling that constrains what future models can understand.

We test this hypothesis by fine-tuning GPT-2 [Radford et al., 2019] on WikiText-103 [Merity et al., 2017] with controlled synthetic contamination across multiple generations, replicated with $n=3$ random seeds. Our contributions are:

1. We introduce the *log-probability gap*, a continuous metric that measures the differential between a model’s processing of standard versus novel text, and its *critical-span* variant that isolates the novel construction tokens from surrounding context.
2. We demonstrate dose-dependent gap widening under synthetic contamination with statistical testing ($n=3$ seeds), establishing that the effect is robust to random variation.

3. We decompose the gap into its constituent parts, revealing two distinct mechanisms: distribution *narrowing* at moderate contamination (10–50%) driven by improved standard-text confidence, and genuine competence *loss* at 100% driven by degraded novel-text processing.
4. We show that deeper fine-tuning (5,000 steps) breaks through the pretrained buffer that protects novel-text comprehension under shallow training (500 steps), establishing that pretrained representations are a buffer, not a firewall.
5. We identify a degenerate collapse threshold (perplexity $> 1,400$) beyond which log-probability gap metrics become unreliable, and we characterize the mechanism by which distributional collapse compresses the gap.

The remainder of this paper is organized as follows. Section 2 surveys prior work on model collapse and synthetic data quality. Section 3 describes our experimental setup, metrics, and benchmark. Section 4 presents results under shallow fine-tuning, and Section 5 extends to deep fine-tuning where the pretrained buffer breaks. Section 6 synthesizes the findings and discusses limitations, and Section 7 concludes.

2 Related Work

Model collapse. Shumailov et al. [2024] established the foundational result: generative models trained recursively on their own outputs lose the tails of the original distribution. They demonstrated this across VAEs, GMMs, and LLMs (OPT-125M), showing that model collapse is architecture-general. Their protocol used full data replacement at each generation—the maximally pessimistic scenario.

Self-consuming generative models. Alemohammad et al. [2023] provided a complementary theoretical analysis, showing that self-consuming generative models “go MAD” (Model Autophagy Disorder), with distribution tails progressively lost across generations regardless of the generative model family.

Synthetic data quality. Hataya et al. [2023] asked whether large-scale generative models will corrupt future datasets, finding that even small fractions of synthetic data in training corpora can shift learned distributions. Martínez et al. [2023] demonstrated distribution shift under synthetic augmentation in classification tasks, and Briesch et al. [2023] showed progressive quality degradation in synthetic data pipelines, extending the collapse finding beyond the recursive self-training setting.

Mitigating collapse through data mixing. Gerstgrasser et al. [2024] showed that if real and synthetic data *accumulate* rather than replace each other, collapse can be avoided. Dohmatob et al. [2024] provided statistical formalization: collapse is unavoidable with 100% synthetic training but can be bounded below a threshold when mixing real and synthetic data. Our experiment uses mixed contamination ratios (0–100%) to map the full degradation curve rather than studying only the extremes.

Comprehension versus generation. Prior work has primarily measured model collapse through generation quality: perplexity, output diversity, and qualitative inspection of generated text. Our work shifts the evaluation target from generation to *comprehension*—the model’s ability to process inputs it has never seen, measured through log-probability assignment to held-out novel linguistic constructions. This distinction proves consequential: we find that comprehension and generation degrade on different timescales.

3 Experimental Setup

3.1 Model and Training

We fine-tune GPT-2 (124M parameters) from HuggingFace pretrained weights on a 2M-token subset of WikiText-103. Fine-tuning uses AdamW with cosine learning rate schedule, effective batch size 32 (micro-batch 4 with 8-step gradient accumulation), and context length 256 tokens. We conduct two training regimes:

- **Shallow:** 500 steps at learning rate 5×10^{-5} , simulating light fine-tuning where pretrained representations are largely preserved.
- **Deep:** 5,000 steps at learning rate 5×10^{-5} (primary) or 1×10^{-4} (aggressive), simulating thorough fine-tuning that may overcome pretrained representations.

All conditions are replicated with 3 random seeds (42, 137, 256); results are reported as mean $\pm$ standard deviation. All models start from the same pretrained GPT-2 weights at each generation to isolate the effect of data contamination from weight drift.

3.2 Data Splits

The WikiText-103 corpus is split into training (90%), validation (5%), and test (5%) sets. Validation data is used exclusively for checkpoint selection during training; all reported metrics are computed on the held-out test set to prevent information leakage.

3.3 Generational Loop

For each contamination ratio $r \in \{0\%, 10\%, 25\%, 50\%, 100\%\}$:

- **Generation 0:** Fine-tune GPT-2 on clean WikiText-103.
- **Generation n ($n \geq 1$):** Generate synthetic text from generation $n-1$'s model using nucleus sampling (top- $p = 0.95$, $T = 1.0$). Mix with the base corpus at ratio r . Fine-tune a fresh GPT-2 on the mixed data.

We run 3 generations per arm. The 0% control uses different effective seeds per generation (seed +1000 $\times$ generation) so that the control arm produces genuinely different models with real variance from random batch ordering, rather than deterministic replicas.

3.4 Evaluation Metrics

Log-probability gap (primary metric). For a model M , we compute mean per-token log-probability on two held-out sets: (i) 200 random passages of 128 tokens from the WikiText test split (*standard*), sampled with a fixed seed for reproducibility, and (ii) 51 hand-curated sentences containing novel linguistic constructions with correct completions filled in (*novel*). The gap is:

$$\Delta_{\text{LP}} = \bar{\ell}_{\text{standard}} - \bar{\ell}_{\text{novel}} \quad (1)$$

where $\bar{\ell}$ denotes mean per-token log-probability. Since novel constructions are harder, $\Delta_{\text{LP}} > 0$. A widening gap under contamination indicates that the model is becoming relatively worse at novel text compared to standard text.

Critical-span log-probability gap. A refined variant that computes $\bar{\ell}_{\text{novel}}$ only over the annotated critical span of each benchmark sentence—the tokens containing the novel construction itself—excluding surrounding context that may be conventional. This variant amplifies the contamination signal by focusing on the tokens that are genuinely out-of-distribution.

Validation perplexity. Standard cross-entropy loss on the held-out WikiText test set, reported as $\exp(\text{loss})$.

Distinct-N. Unigram, bigram, and trigram type-to-token ratio over 50 prompted generations of 100 tokens each [Li et al., 2016].

Novelty accuracy. Multiple-choice cloze accuracy on the 51-example linguistic novelty benchmark. Each example presents a sentence with a blank and four options; the model selects the option with the highest mean log-probability over the completion span.

3.5 Linguistic Novelty Benchmark

The benchmark comprises 51 examples across five categories of linguistic phenomena that post-date or are absent from GPT-2’s training distribution: novel slang (20 examples), creative neologisms (14), rhetorical subversion (8), pragmatic inference (6), and code-switching (3). Each example includes character-level annotations of the critical span containing the novel construction, enabling the critical-span gap variant.

3.6 Compute

All experiments are run on a single Apple M4 Pro (24 GB unified memory) using PyTorch’s MPS backend. Table 1 summarizes the compute budget. Total wall-clock time across all experiments is approximately 75 hours, dominated by the deep fine-tuning multi-seed sweep.

Table 1: Compute budget. All runs on Apple M4 Pro, 24 GB RAM, MPS backend.

Phase	Runs	Time per run	Total
Shallow (V2) multi-seed	54	~25 min	14.5 h
Deep (V4) multi-seed	45 (30 trained)	~2.5 h	60.2 h
V3 critical-span re-eval	45	<1 min	~0.5 h
Total			~75 h

4 Results: Shallow Fine-Tuning

4.1 Log-Probability Gap Widens with Contamination

Table 2 presents the log-probability gap across all shallow fine-tuning conditions. The 0% control arm produces a stable gap of 0.49 ± 0.01 across all generations. Contaminated arms show statistically significant dose-dependent gap widening at generation 1, with the 100% arm producing a gap of 1.36 ± 0.10 —nearly $3\times$ the baseline. Figure 1 visualizes this trend with error bands across seeds.

4.2 Decomposing the Gap: Distribution Narrowing

Table 3 decomposes the gap into its standard and novel components at generation 1. At 10–50% contamination, the gap widens primarily because standard-text log-probability *improves* (from -5.72 toward -5.46) while novel-text log-probability remains approximately constant (~ -6.21). The model becomes more confident on in-distribution text without losing capacity for novel constructions—distribution *narrowing* rather than competence loss.

Table 2: Log-probability gap (Δ_{LP} , mean $\pm$ std, $n=3$) under shallow fine-tuning (500 steps). Higher values indicate a wider differential between standard and novel text processing.

Contamination	Gen 0	Gen 1	Gen 2
0% (control)	0.490 ± 0.002	0.504 ± 0.017	0.492 ± 0.012
10%	0.490 ± 0.002	0.604 ± 0.012	0.589 ± 0.002
25%	0.490 ± 0.002	0.643 ± 0.014	0.614 ± 0.003
50%	0.490 ± 0.002	0.745 ± 0.011	0.660 ± 0.008
100%	0.490 ± 0.002	1.363 ± 0.103	1.040 ± 0.130
Compound (25%)	0.490 ± 0.002	0.643 ± 0.014	0.615 ± 0.001

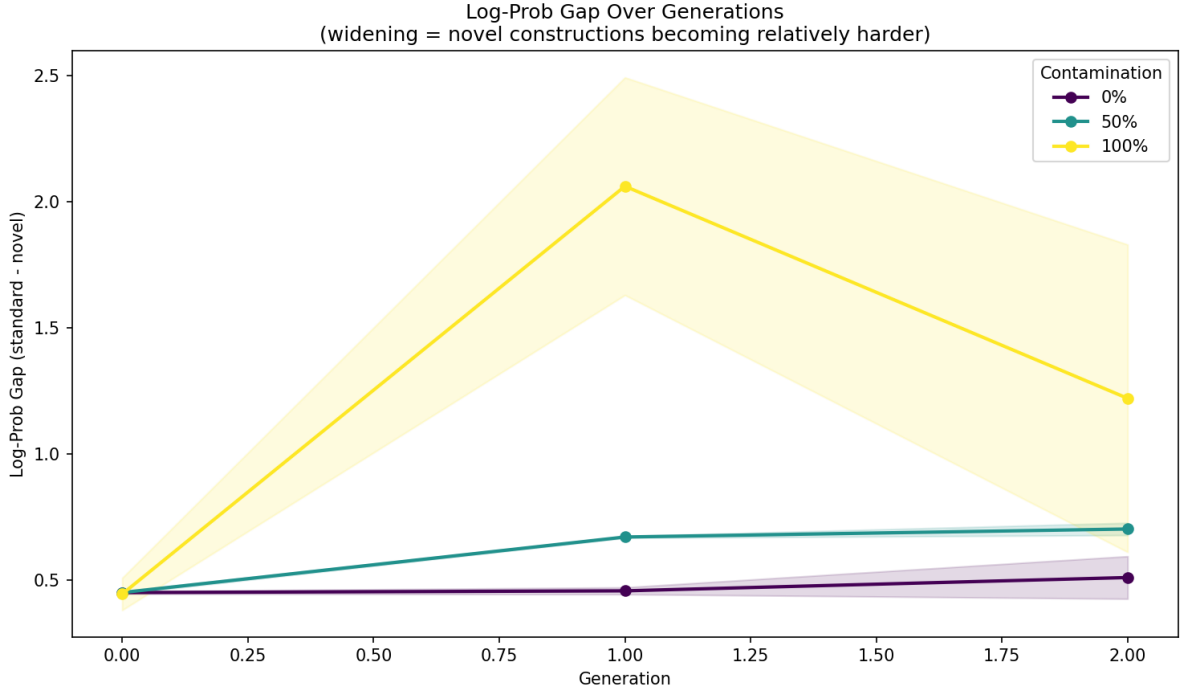


Figure 1: Log-probability gap across generations and contamination ratios under shallow fine-tuning. Error bands show ± 1 standard deviation over $n=3$ seeds. The dose-dependent widening is monotonic at generation 1; partial recovery at generation 2 reflects the degenerate collapse artifact at 100% contamination (Section 5.6).

At 100% contamination, both mechanisms contribute: standard-text log-probability improves sharply ($-5.72 \rightarrow -5.08$) *and* novel-text log-probability degrades ($-6.21 \rightarrow -6.44$). This is the only condition under shallow training where novel-text processing genuinely worsens.

Table 3: Decomposition of mean log-probability at generation 1 (shallow, mean $\pm$ std, $n=3$). At moderate contamination, the gap widens because standard text gets easier; at 100%, novel text also gets harder.

Contamination	$\bar{\ell}_{\text{std}}$ (Gen 0)	$\bar{\ell}_{\text{std}}$ (Gen 1)	$\bar{\ell}_{\text{nov}}$ (Gen 0)	$\bar{\ell}_{\text{nov}}$ (Gen 1)
0%	-5.72 ± 0.01	-5.71 ± 0.03	-6.21 ± 0.01	-6.21 ± 0.03
10%	-5.72 ± 0.01	-5.62 ± 0.01	-6.21 ± 0.01	-6.22 ± 0.02
25%	-5.72 ± 0.01	-5.57 ± 0.01	-6.21 ± 0.01	-6.21 ± 0.02
50%	-5.72 ± 0.01	-5.46 ± 0.01	-6.21 ± 0.01	-6.20 ± 0.02
100%	-5.72 ± 0.01	-5.08 ± 0.14	-6.21 ± 0.01	-6.44 ± 0.25

4.3 Validation Perplexity

Perplexity degrades monotonically with contamination (Table 4), replicating the model collapse findings of Shumailov et al. [2024]. At 10–50% contamination, the degradation is modest (5–22%). The 100% arm exhibits catastrophic collapse, reaching a perplexity of $1,697 \pm 112$ by generation 2.

Table 4: Validation perplexity (mean $\pm$ std, $n=3$) under shallow fine-tuning. The 100% arm shows catastrophic collapse; moderate contamination produces modest degradation.

Contamination	Gen 0	Gen 1	Gen 2
0%	247 ± 2	248 ± 4	248 ± 4
10%	247 ± 2	261 ± 2	261 ± 2
25%	247 ± 2	274 ± 2	273 ± 2
50%	247 ± 2	303 ± 2	300 ± 3
100%	247 ± 2	954 ± 12	$1,697 \pm 112$

4.4 Generation Diversity Collapses

Distinct-1 (unigram diversity) provides the clearest experimental signal (Table 5 and Figure 2). Even 10% contamination reduces diversity from 0.43 to 0.27—a 37% drop—while the log-probability gap at the same condition shows only a modest increase from 0.49 to 0.60. This asymmetry between generation and comprehension degradation is a central finding: synthetic contamination first destroys generative diversity before affecting comprehension. At 100% contamination by generation 2, Distinct-1 falls to 0.04, indicating near-repetitive output. The 0% control remains stable across all generations.

4.5 Novelty Accuracy and Compounding

Multiple-choice novelty accuracy remained flat across all conditions (0.29–0.37), confirming that GPT-2 answers these forced-choice questions from token-frequency priors rather than pragmatic comprehension. This metric is too coarse to detect the effects captured by the continuous log-probability gap.

The compounding arm (25% contamination with model-weight inheritance across generations) produced results nearly identical to the fixed 25% arm: a generation-1 gap of 0.643 versus 0.643, and a generation-2 gap of 0.615 versus 0.614. Over 3 generations, compounding confers

Table 5: Distinct-1 (mean $\pm$ std, $n=3$) under shallow fine-tuning. Diversity collapses at all contamination levels, providing the earliest signal of distribution narrowing.

Contamination	Gen 0	Gen 1	Gen 2
0%	0.426 ± 0.006	0.438 ± 0.006	0.441 ± 0.008
10%	0.428 ± 0.006	0.272 ± 0.008	0.267 ± 0.013
25%	0.428 ± 0.003	0.226 ± 0.027	0.182 ± 0.014
50%	0.429 ± 0.003	0.203 ± 0.010	0.164 ± 0.004
100%	0.431 ± 0.007	0.103 ± 0.004	0.042 ± 0.004

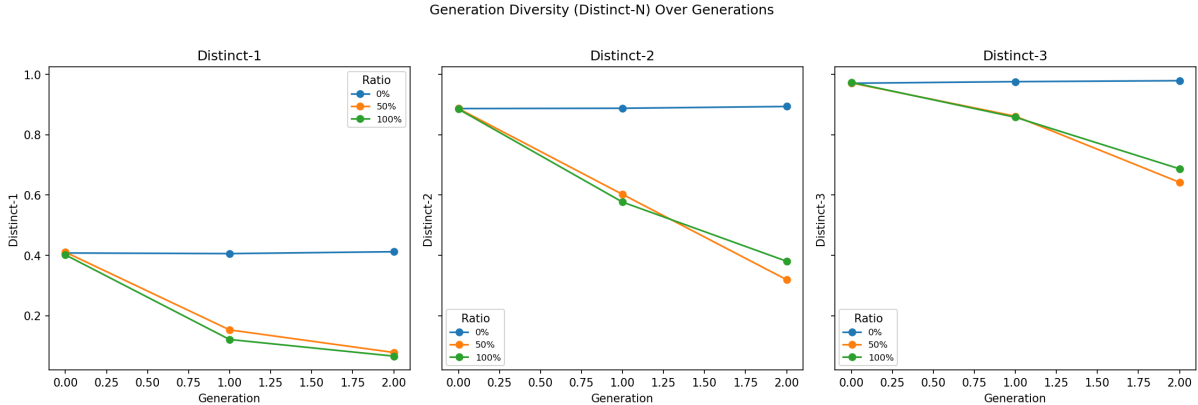


Figure 2: Distinct-N diversity metrics across generations and contamination ratios. Diversity collapses sharply even at 10% contamination, making it the most sensitive early-warning signal of distribution narrowing.

no additional degradation beyond what fixed-ratio contamination produces, suggesting rapid saturation to a contaminated equilibrium.

5 Results: Deep Fine-Tuning Breaks the Pretrained Buffer

The shallow fine-tuning results raised a critical question: is the preservation of novel-text comprehension at moderate contamination a genuine property of the data, or merely an artifact of insufficient optimization? GPT-2’s pretrained representations may act as a buffer that resists contamination under 500 steps of fine-tuning but would yield under more aggressive training. We test this hypothesis by extending fine-tuning to 5,000 steps ($10\times$ longer) at contamination ratios of 0%, 50%, and 100%.

5.1 The Gap Under Deep Training

Table 6 presents the deep fine-tuning results at learning rate 5×10^{-5} . The key finding is at 100% contamination, generation 1: the gap widens to 2.30 ± 0.09 ($p < 0.001$ vs. control), compared to 1.36 ± 0.10 under shallow training. Deeper optimization drives the model further from its pretrained representations, amplifying the contamination signal by $1.7\times$.

At 50% contamination, the gap also widens significantly (0.67 ± 0.004 , $p < 0.001$), and unlike the shallow regime, the gap continues to grow at generation 2 (0.70 ± 0.02 , $p = 0.036$) rather than recovering.

Table 6: Log-probability gap (Δ_{LP} , mean $\pm$ std) under deep fine-tuning (5,000 steps, $\text{LR} = 5 \times 10^{-5}$, $n=3$). Asterisks denote significance vs. 0% control (* $p < 0.05$, ** $p < 0.001$).

Contamination	Gen 0	Gen 1	Gen 2
0% (control)	0.449 ± 0.005	0.456 ± 0.014	0.509 ± 0.084
50%	0.449 ± 0.005	$0.669 \pm 0.004^{**}$	$0.701 \pm 0.024^*$
100%	0.449 ± 0.005	$2.297 \pm 0.091^{**}$	$0.917 \pm 0.021^\dagger$

† Not significant ($p = 0.12$). See Section 5.6 for discussion of the degenerate collapse artifact.

5.2 Perplexity Confirms Monotonic Collapse

While the log-probability gap shows an apparent “recovery” at 100% generation 2, perplexity tells the true story (Table 7): monotonic degradation from $192 \rightarrow 854 \rightarrow 1,482$. The gap compression at generation 2 is a metric artifact, not model improvement (Section 5.6). Figure 3 visualizes the relationship between perplexity and gap across all conditions, revealing the degenerate region where the two metrics diverge.

Table 7: Validation perplexity (mean $\pm$ std, $n=3$) under deep fine-tuning. Monotonic collapse at 100% contamination confirms progressive model degradation despite apparent gap “recovery.”

Contamination	Gen 0	Gen 1	Gen 2
0%	192 ± 0	192 ± 0	192 ± 1
50%	192 ± 0	233 ± 1	236 ± 0
100%	192 ± 0	854 ± 23	$1,482 \pm 30$

5.3 Aggressive Learning Rate Amplifies Degradation

To further test whether deeper optimization breaks the pretrained buffer, we repeated the 100% arm with a $2\times$ higher learning rate (1×10^{-4}). Table 8 shows qualitatively similar results: the gap rises to 1.82 ± 0.50 at generation 1 and 1.52 ± 0.75 at generation 2, with perplexity reaching $1,603 \pm 57$. The higher variance reflects the instability of aggressive optimization on contaminated data. Figure 4 compares the two learning rate configurations directly.

Table 8: Aggressive learning rate (1×10^{-4}) at 100% contamination (mean $\pm$ std, $n=3$). Higher LR produces qualitatively similar degradation with greater variance.

Gen	Δ_{LP}	$\Delta_{\text{LP}}^{\text{crit}}$	PPL	D-1
0	0.437 ± 0.090	1.147 ± 0.105	204 ± 10	0.391 ± 0.006
1	1.825 ± 0.503	2.797 ± 0.704	845 ± 33	0.122 ± 0.017
2	1.521 ± 0.749	2.283 ± 0.917	$1,603 \pm 57$	0.054 ± 0.008

5.4 Decomposition: Deep Training Degrades Novel-Text Processing

The critical difference between shallow and deep fine-tuning is visible in the decomposition (Table 9 and Figure 5). Under deep training at 100% contamination generation 1, novel-text log-probability drops to -8.09 ± 0.06 , compared to -6.44 ± 0.25 under shallow training—a 1.65 nats further decline. The pretrained buffer that preserved novel-text processing under shallow training has been overcome: 5,000 steps of optimization on contaminated data is sufficient to degrade the model’s internal representations of out-of-distribution language.

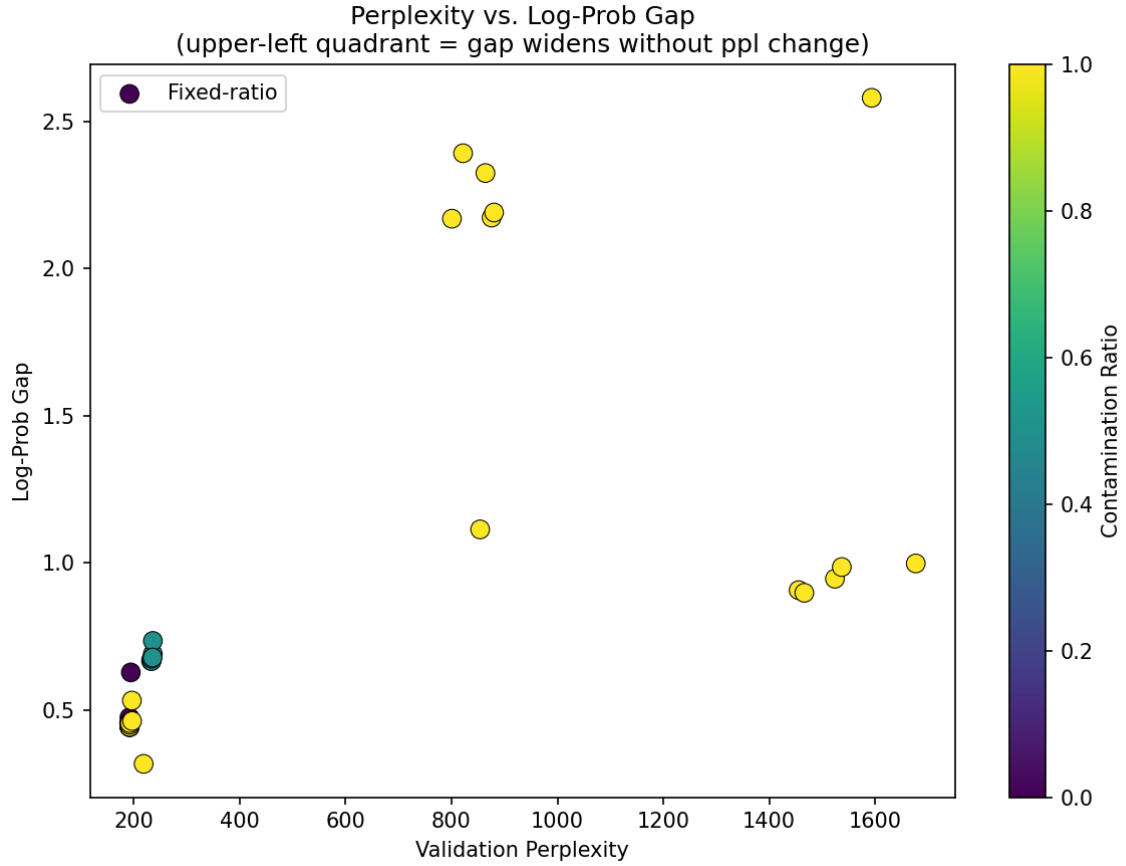


Table 9: Decomposition at generation 1 under deep fine-tuning (primary LR, mean $\pm$ std, $n=3$). Novel-text log-probability degrades sharply at 100%, unlike under shallow training where it remained stable at moderate contamination.

Contamination	$\bar{\ell}_{\text{std}}$ (Gen 0)	$\bar{\ell}_{\text{std}}$ (Gen 1)	$\bar{\ell}_{\text{nov}}$ (Gen 0)	$\bar{\ell}_{\text{nov}}$ (Gen 1)
0%	-6.71 ± 0.02	-6.71 ± 0.00	-7.16 ± 0.02	-7.17 ± 0.01
50%	-6.71 ± 0.02	-6.42 ± 0.02	-7.16 ± 0.02	-7.09 ± 0.03
100%	-6.71 ± 0.02	-5.79 ± 0.10	-7.16 ± 0.02	-8.09 ± 0.06

V4: Decomposed Log-Prob by Config

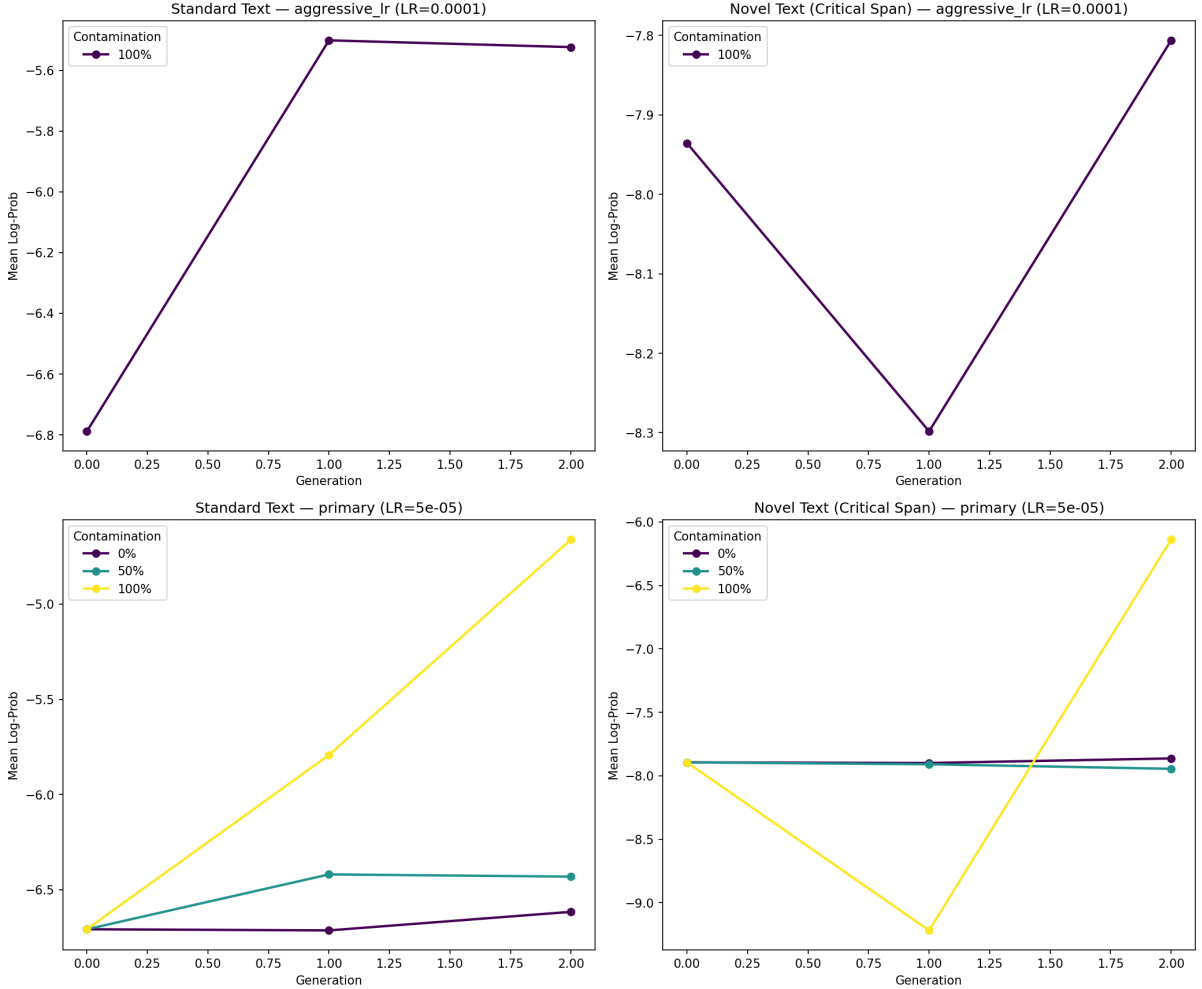


Figure 5: Decomposed log-probability components under deep fine-tuning. Standard-text log-probability (dashed) improves at all contamination levels; novel-text log-probability (solid) degrades sharply at 100%, revealing that the pretrained buffer has been overcome.

5.5 Critical-Span Gap Amplifies the Signal

The critical-span variant of the gap, which scores only the novel construction tokens rather than the full sentence, consistently amplifies the contamination signal (Figures 6 and 7). Under deep training at 100% contamination generation 1, the critical-span gap reaches 3.43 ± 0.11 compared to 2.30 ± 0.09 for the full-sentence gap. This confirms that the novel construction tokens themselves—not the surrounding conventional context—are the locus of degradation.

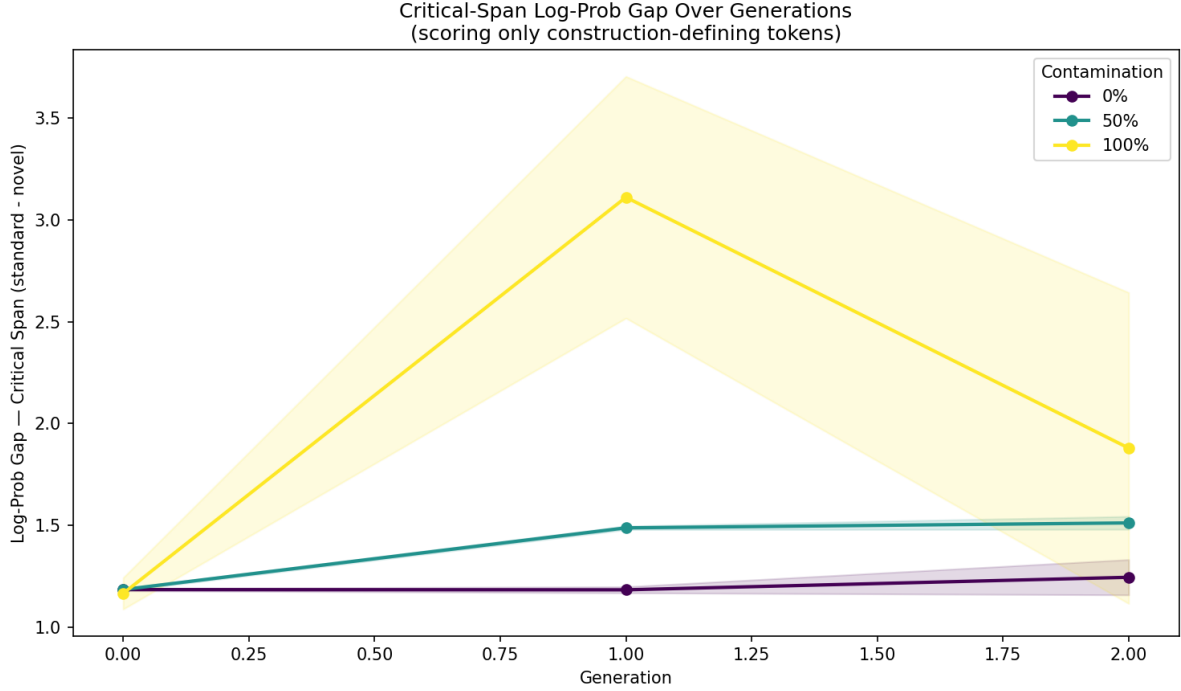


Figure 6: Critical-span log-probability gap, measuring only the tokens within annotated novel construction spans. The contamination signal is amplified relative to the full-sentence gap (Figure 1).

5.6 The Degenerate Collapse Artifact

At 100% contamination generation 2 under deep training, the primary configuration shows an apparent “recovery” in the log-probability gap (0.92 ± 0.02), despite perplexity reaching $1,482 \pm 30$. This is a metric artifact, not model improvement.

The mechanism is as follows. By generation 2, the model’s probability distribution has collapsed to a narrow spike over a small set of memorized patterns. Standard-text log-probability jumps from its baseline of -6.71 to -4.66 : the model assigns absurdly high confidence to the few patterns it has memorized. Novel-text log-probability is also pulled upward (from -7.16 to -5.58) as both scores compress toward the distorted distribution. The gap shrinks because both components are pulled toward the same degenerate attractor, not because the model has recovered any competence.

We identify perplexity $> 1,400$ as a practical threshold for degenerate collapse, beyond which log-probability gap metrics should not be interpreted at face value. Perplexity itself remains a reliable monotonic indicator of degradation throughout.

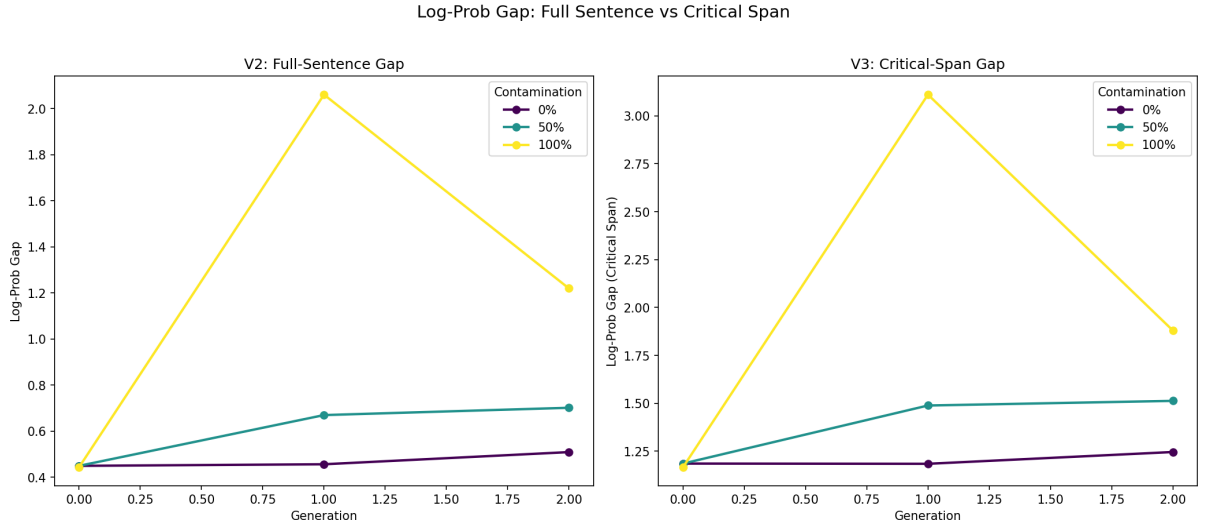


Figure 7: Comparison of full-sentence and critical-span gap variants. The critical-span gap (right) is uniformly larger and shows clearer separation between contamination levels, confirming that novel construction tokens are the primary locus of degradation.

6 Discussion

6.1 Training Dynamics

Figure 8 shows validation loss curves for generation 1 models at different contamination levels. Under shallow training, all conditions converge within 500 steps, with contaminated arms settling to higher loss values. Under deep training, the 0% and 50% arms converge smoothly, but the 100% arm exhibits persistently elevated loss throughout 5,000 steps—the model never recovers from the distribution mismatch introduced by fully synthetic training data. The aggressive learning rate configuration at 100% shows even higher loss, confirming that stronger optimization on contaminated data worsens rather than overcomes the distribution shift.

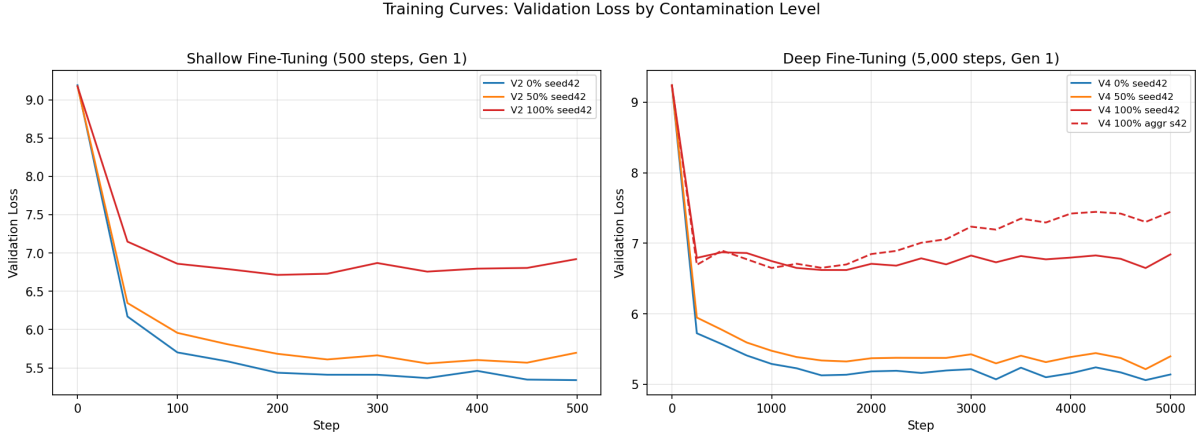


Figure 8: Validation loss curves for generation 1 models under shallow (left) and deep (right) fine-tuning. Under deep training, the 100% contamination arms settle to substantially higher loss, reflecting persistent distribution mismatch from fully synthetic data.

6.2 Two Regimes of Degradation

Our results reveal two distinct contamination regimes, mediated by optimization depth:

1. **Shallow training + moderate contamination:** The model develops a narrower but more confident distribution over standard text. Novel constructions are not degraded; they are left behind as the model specializes. This is distribution *narrowing*—the model becomes a specialist rather than a generalist—but not competence loss.
2. **Deep training + high contamination:** The pretrained buffer breaks. Novel-text log-probabilities degrade sharply ($-7.16 \rightarrow -8.09$), and perplexity collapses into the degenerate regime. This is genuine competence *loss*: the model’s internal representations of out-of-distribution language are overwritten by optimization on contaminated data.

The practical implication is that pretrained representations offer meaningful but *not unlimited* protection against synthetic contamination. Light fine-tuning on moderately contaminated data may be safe; thorough fine-tuning on heavily contaminated data is not.

6.3 Diversity Collapse as an Early Warning

Generation diversity proved to be the most sensitive indicator of distribution narrowing across both training regimes, collapsing even at 10% contamination under shallow training while the log-probability gap at the same condition showed only modest movement. Distinct-N monitoring could serve as a practical early-warning signal for synthetic contamination in training pipelines:

a sharp drop relative to the pretrained baseline may indicate distributional narrowing before it manifests in comprehension metrics or downstream task performance.

6.4 Comprehension vs. Generation: An Asymmetry

A central finding is the asymmetry between generation and comprehension degradation. Under shallow training at 10% contamination, Distinct-1 drops 37% while the log-probability gap increases by only 23%. This suggests that synthetic contamination first destroys generative diversity—the model’s ability to *produce* varied output—before affecting comprehension—its ability to *process* varied input. The asymmetry narrows under deep training, where both generation and comprehension degrade together, consistent with the pretrained buffer breaking down.

6.5 Limitations

1. **Small model.** GPT-2 124M may not be representative of larger models, which have deeper and more robust pretrained representations. The buffer may be substantially more durable in models with billions of parameters.
2. **Only 3 generations.** Shumailov et al. [2024] used up to 10 generations. Additional generations may reveal whether moderate contamination eventually breaks the buffer even under shallow training.
3. **Self-contamination only.** All synthetic data is generated by variants of the same architecture. Cross-model contamination (e.g., training GPT-2 on data generated by a different model family) might produce different dynamics.
4. **Benchmark size.** With 51 examples across 5 categories, the benchmark provides limited statistical power for per-category analysis. Categories with 3–6 items (code-switching, pragmatic inference) cannot support reliable subcategory conclusions. We report a preliminary per-category breakdown in Section A but caution against strong claims from small- n categories.
5. **Token-level metric.** The log-probability gap measures token prediction rather than pragmatic comprehension directly. A model that assigns low probability to novel tokens may still “comprehend” the construction in some functional sense not captured by this metric.

7 Conclusion

We tested whether synthetic data contamination destroys a language model’s ability to process novel linguistic constructions, using multi-seed replication ($n=3$) across two training depths.

Under shallow fine-tuning, moderate contamination (10–50%) produces distribution *narrowing*: the model specializes toward conventional text without losing capacity for novel constructions. Under deep fine-tuning, the pretrained buffer breaks: at 100% contamination, the log-probability gap widens to 2.30 ± 0.09 ($p < 0.001$) with perplexity collapsing from 192 to 1,482 across two generations of recursive training.

These findings establish that pretrained representations are a buffer, not a firewall. They protect novel-text comprehension under light training but yield under sufficient optimization pressure on contaminated data. Generation diversity emerges as the earliest and most sensitive indicator of distributional narrowing, collapsing at contamination levels where comprehension metrics show no effect.

We recommend Distinct-N monitoring as a practical diagnostic for synthetic contamination in training pipelines, and we caution against interpreting log-probability gap metrics when model

perplexity exceeds the degenerate collapse threshold ($\text{PPL} > 1,400$). Future work should extend this analysis to larger models with deeper pretrained buffers, more generations of recursive training, and a finer contamination sweep in the 50–100% range to precisely locate the threshold at which narrowing transitions to competence loss.

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Appendix

A Per-Category Sensitivity Analysis

To investigate whether certain types of novel constructions are more vulnerable to contamination than others, we compute the critical-span gap change from baseline (0% contamination, generation 0) to the most degraded non-degenerate condition (100% contamination, generation 1) under deep fine-tuning. Table 10 reports the results per category, and Figures 9 and 10 visualize the patterns.

All five categories show substantial degradation, with critical-span gaps increasing by 160–260% relative to baseline. Rhetorical subversion shows the largest relative increase (+262%), suggesting that pragmatically complex constructions—where meaning depends on subverting the expected register or tone—are particularly fragile under contamination. Creative neologisms show the smallest relative increase (+158%), possibly because novel word forms are already poorly modeled at baseline, leaving less room for further degradation.

These results should be interpreted cautiously given the small per-category sample sizes, particularly for code-switching ($n=3$) and pragmatic inference ($n=6$).

Table 10: Per-category critical-span gap sensitivity under deep fine-tuning (primary LR, mean over $n=3$ seeds). Δ is the change from baseline (0%, Gen 0) to contaminated (100%, Gen 1).

Category	N	Gap (0%, G0)	Gap (100%, G1)	Δ	Rel.
Rhetorical subversion	8	0.904	3.270	+2.366	+262%
Code-switching	3	1.101	3.470	+2.368	+215%
Pragmatic inference	6	1.270	3.565	+2.296	+181%
Novel slang	20	1.333	3.586	+2.253	+169%
Creative neologism	14	1.242	3.203	+1.961	+158%

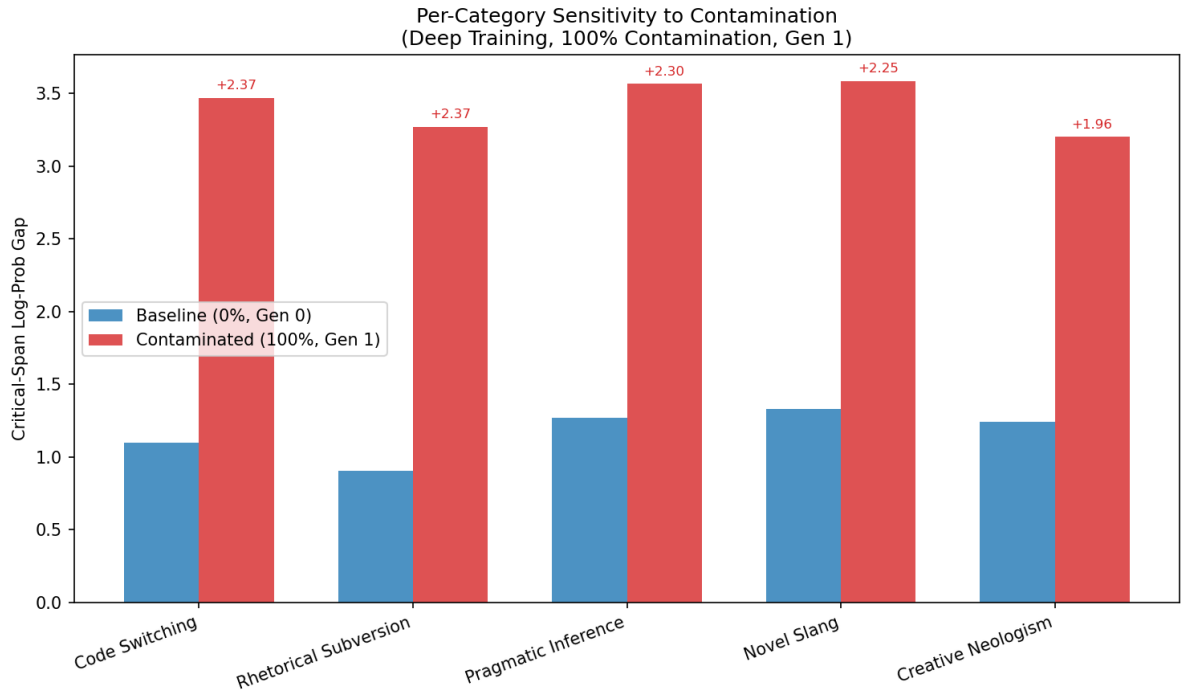


Figure 9: Per-category critical-span gap at baseline vs. 100% contamination generation 1 under deep fine-tuning. All categories show substantial degradation; annotations show the absolute gap increase.

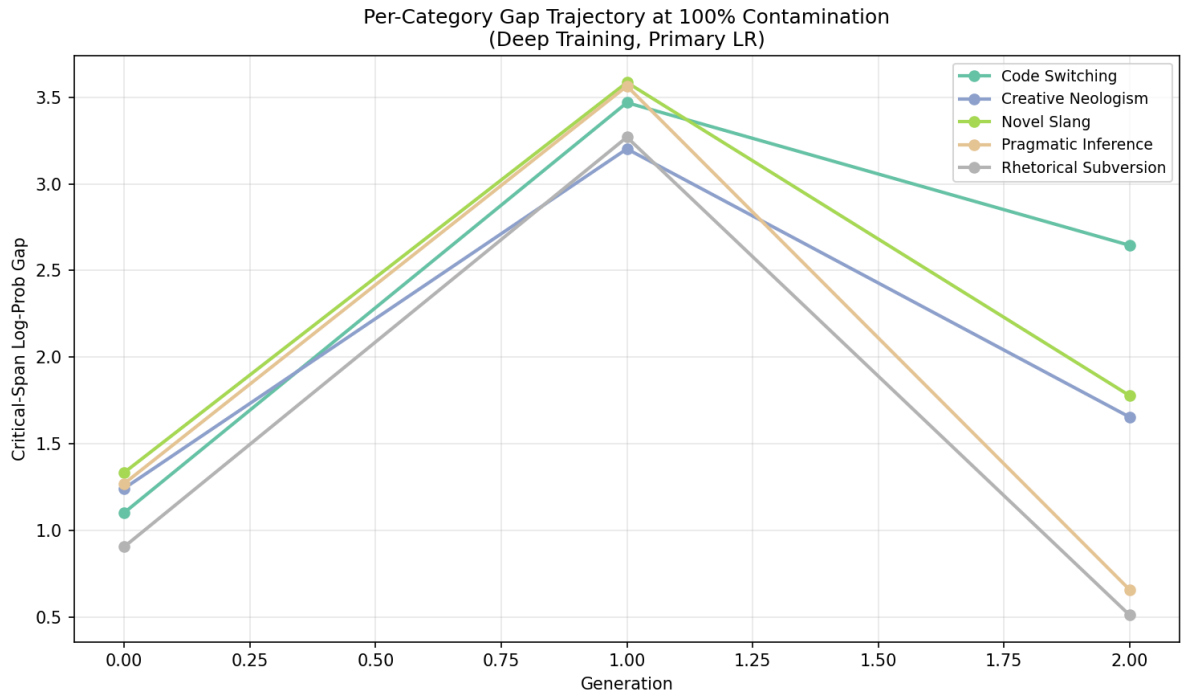


Figure 10: Per-category critical-span gap trajectory across generations at 100% contamination (deep fine-tuning, primary LR). All categories peak at generation 1 and compress at generation 2 due to the degenerate collapse artifact.